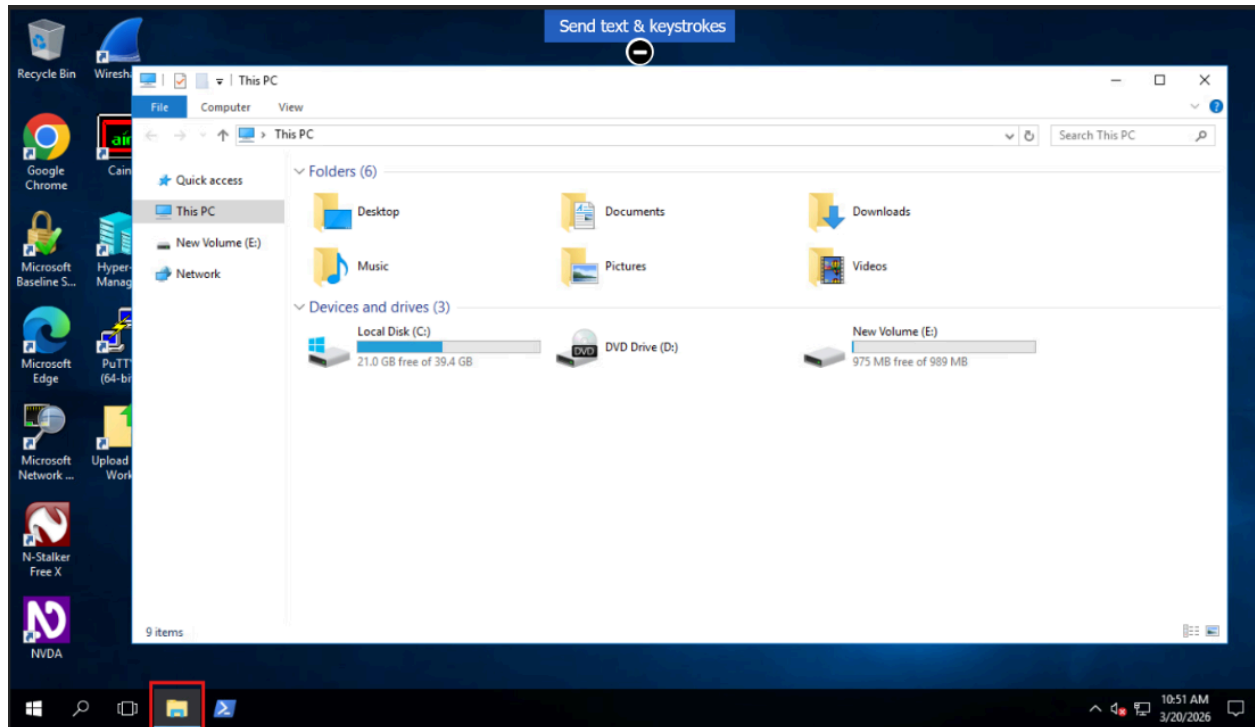
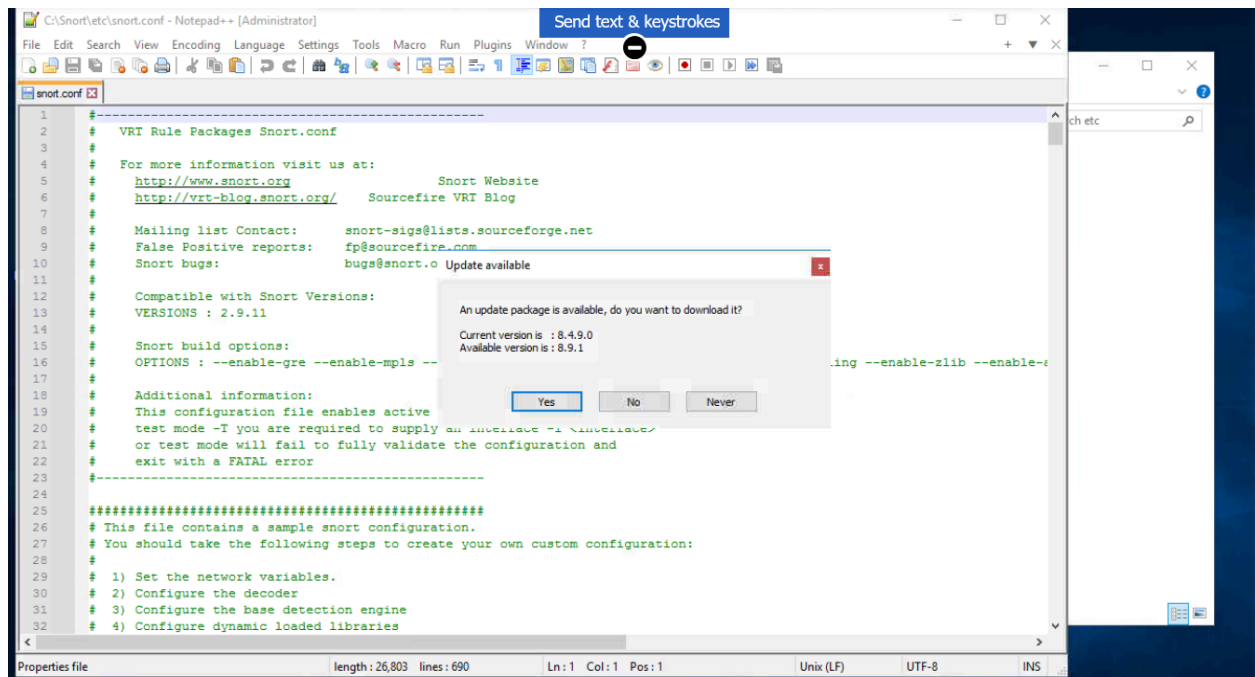
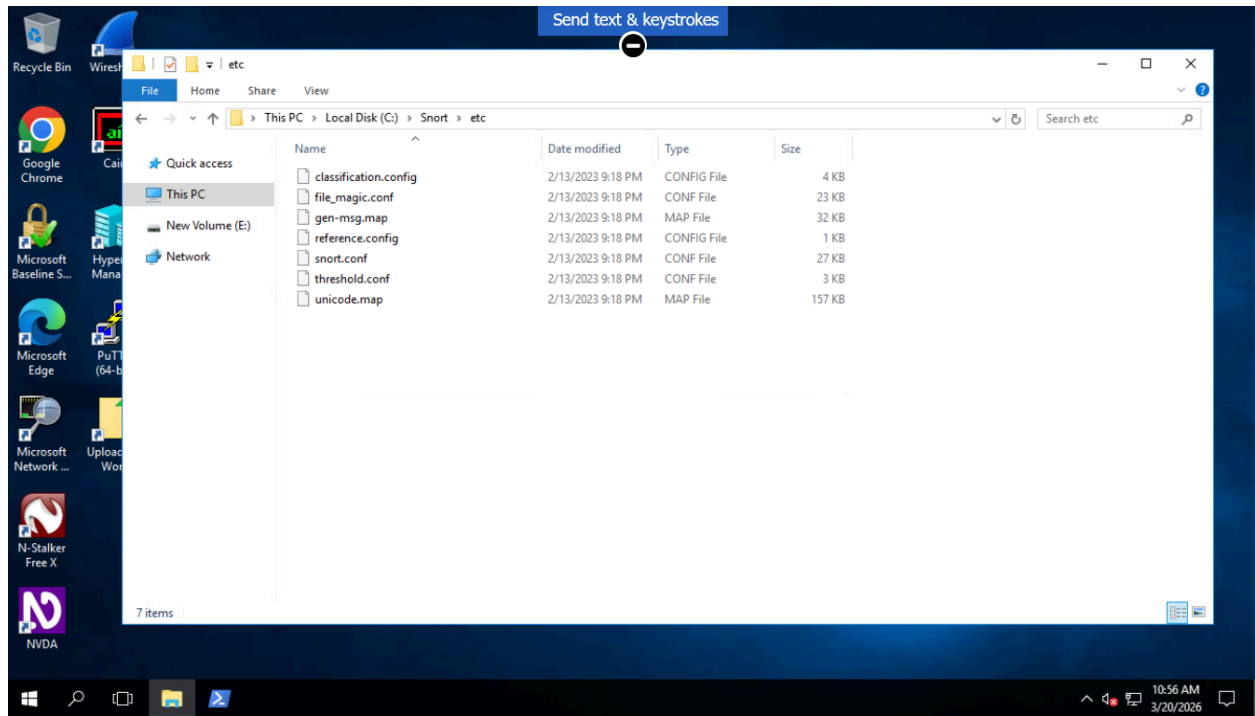


In this lab, you learn to configure snort. Snort is a network monitoring tool used as an IDS and works on defined signatures. To configure snort. Here's what you need to do:

1. From the taskbar, open **File Explorer**.



2. Navigate to **Local Disk (C:) > Snort > etc**, right-click **snort.conf**, and then select **Edit with Notepad++**.



3. Maximize the **Notepad** window.
4. In the **snort.conf** file, change the following things:

Comment line **99**.

In line **104**, change the **RULE_PATH** as *C:\Snort\rules*

Comment line **105**.

In line **106**, set **PREPROC_RULE_PATH** as *C:\Snort\preproc_rules*

In line **113**, set **WHITE_LIST_PATH** as *C:\Snort\rules*

In line **114**, set **BLACK_LIST_PATH** as *C:\Snort\rules*

```
# List of ports you run SIP servers on
portvar SIP_PORTS [5060,5061,5600]

# List of file data ports for file inspection
portvar FILE_DATA_PORTS [$HTTP_PORTS,110,143]

# List of GTP ports for GTP preprocessor
portvar GTP_PORTS [2123,2152,3386]

# other variables, these should not be modified
# ipvar AIM_SERVERS [64.12.24.0/23,64.12.28.0/23,64.12.161.0/24,64.12.163.0/24,64.12.200.0/24,205.188.3.0/24,205.188.5.0/24,205.188.7.0/24,205.188.9.0/24]

# Path to your rules files (this can be a relative path)
# Note for Windows users: You are advised to make this an absolute path,
# such as: c:\snort\rules
var RULE_PATH C:\Snort\rules
#var SO_RULE_PATH ../so_rules
var PREPROC_RULE_PATH C:\Snort\preproc_rules

# If you are using reputation preprocessor set these
# Currently there is a bug with relative paths, they are relative to where snort is
# not relative to snort.conf like the above variables
# This is completely inconsistent with how other vars work, BUG 89986
# Set the absolute path appropriately
var WHITE_LIST_PATH C:\Snort\rules
var BLACK_LIST_PATH C:\Snort\rules
|
*****
```

Uncomment line **186**, and set **config logdir** as *C:\Snort\log*

```
186 | config logdir: C:\Snort\log
```

In line **247**, set **dynamicpreprocessor directory** as

C:\Snort\lib\snort_dynamicpreprocessor

In line **250**, set **dynamicengine** as

```
246 | # path to dynamic preprocessor libraries
247 | dynamicpreprocessor directory C:\Snort\lib\snort_dynamicpreprocessor
248 |
249 | # path to base preprocessor engine
250 | dynamicengine C:\Snort\lib\snort_dynamicengine\sf_engine.dll
251 |
252 | # path to dynamic engine libraries
```

C:\Snort\lib\snort_dynamicengine\sf_engine.dll

Comment lines **253**, **265**, **266**, **267**, **268**, **269**, **335**.

Uncomment line **418**.

In line **511**, set **whitelist** as *\$WHITE_LIST_PATH\white.list, *

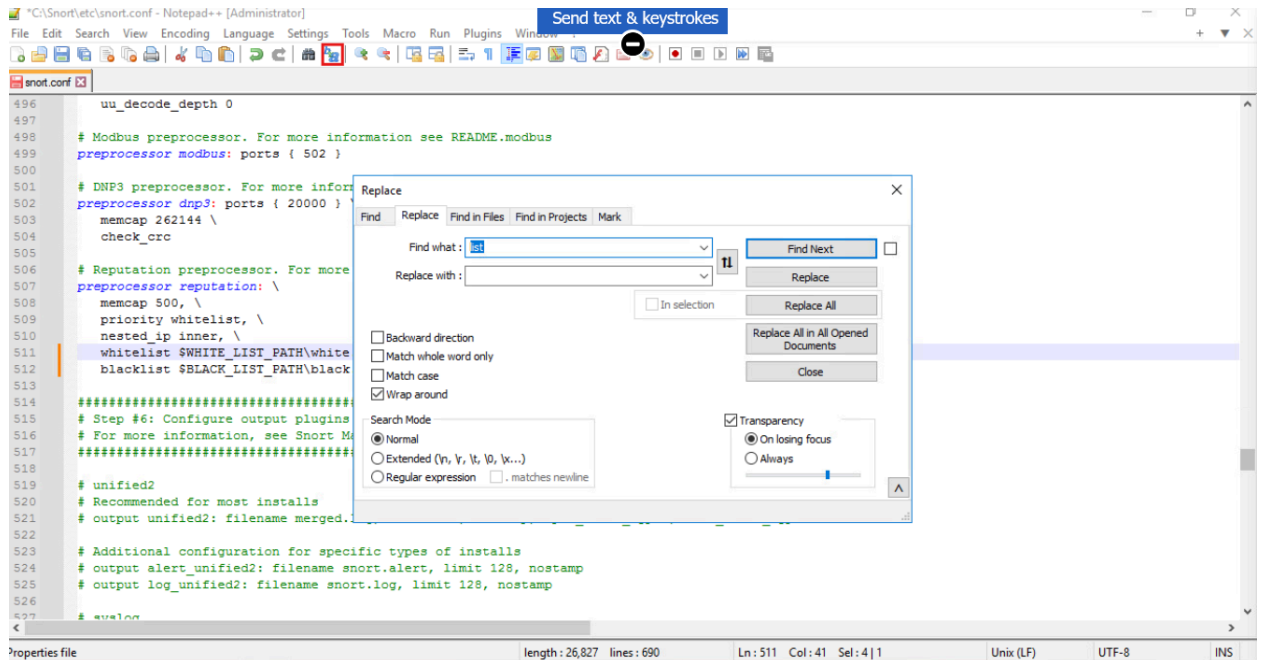
In line **512**, set **blacklist** as *\$BLACK_LIST_PATH\black.list*

```
511 whitelist $WHITE_LIST_PATH\white.list,, \
512 blacklist $BLACK_LIST_PATH\black.list
```

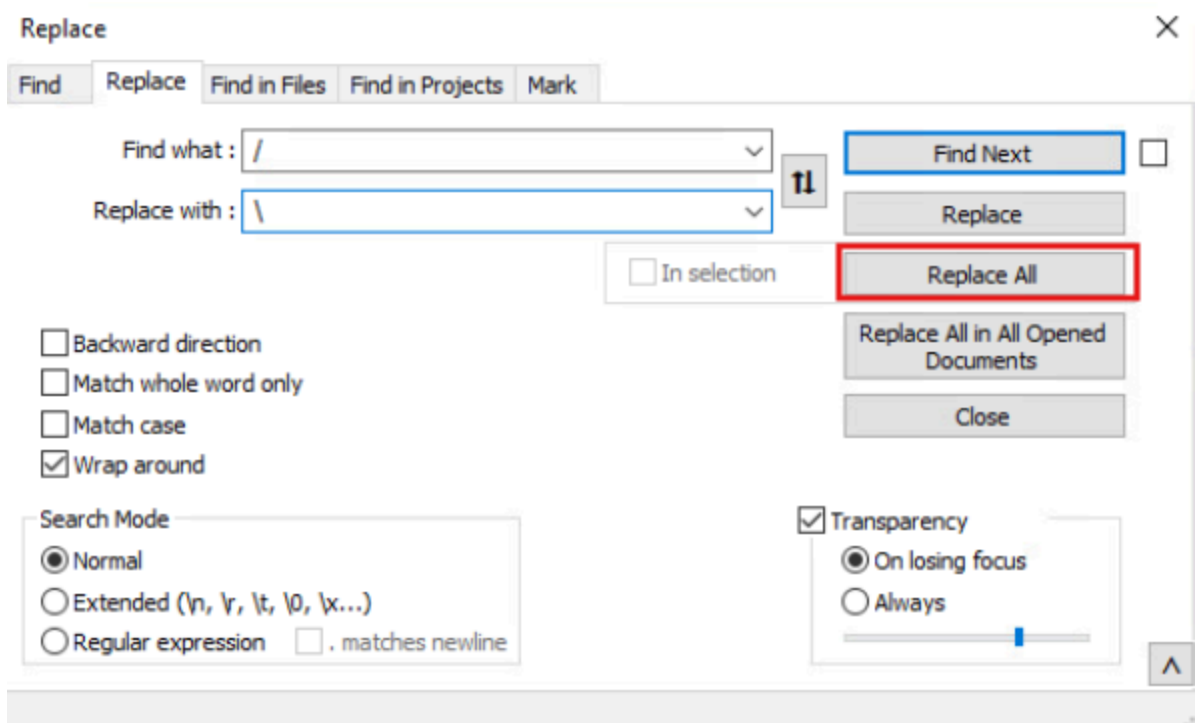
On the icon-based toolbar, select the **Replace** (



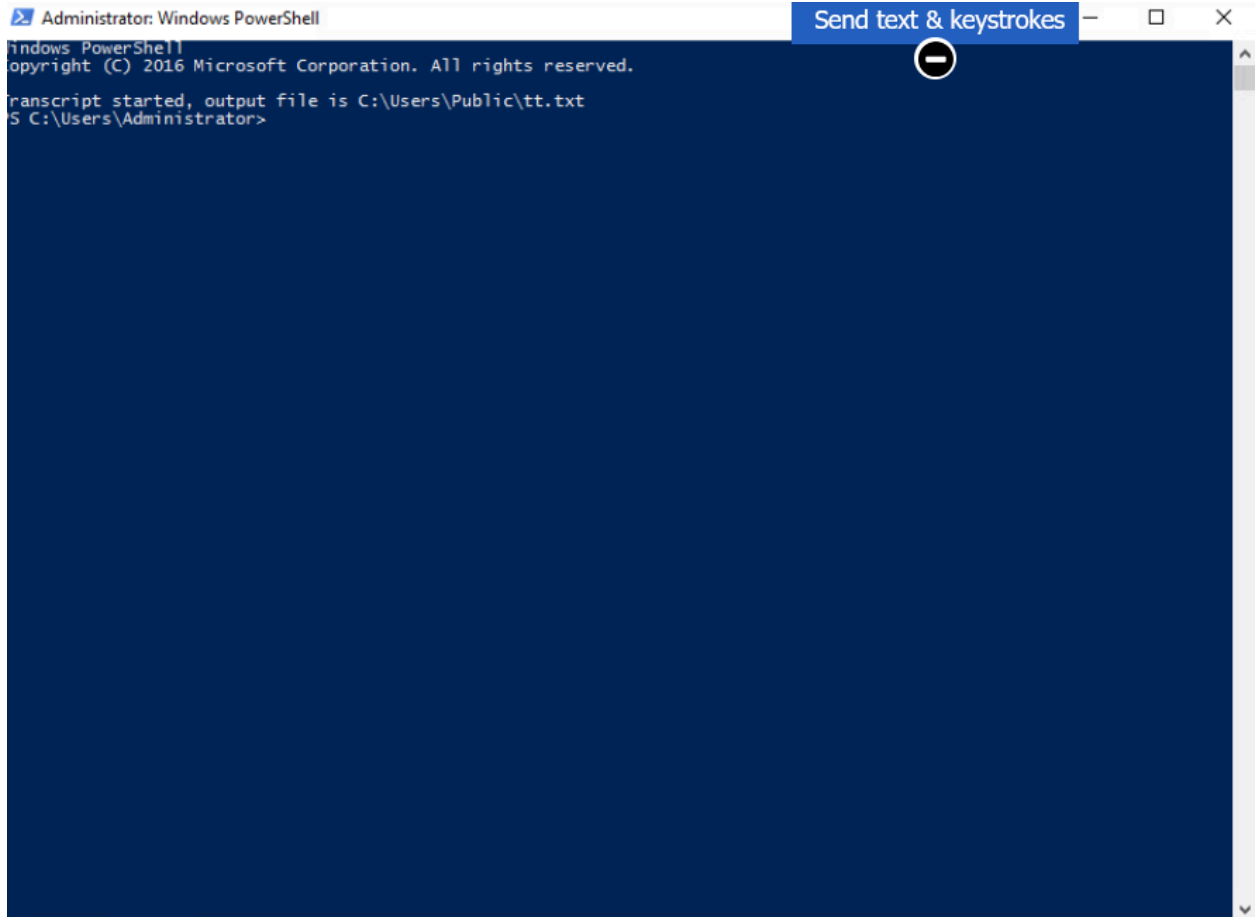
) icon to open the **Replace** dialog box.



In the **Replace** dialog box, type **Find what** as **/**, **Replace with** as ****, and click **Replace All**.



5. Close the **Replace** dialog box.
6. From the menu, navigate to **File > Save** to save all the changes.
7. On the taskbar, click the **Start** () icon and open **Windows PowerShell**.



8. Execute the following commands one by one:

```
cd \snort\bin  
.\snort.exe -V -i 0
```

cd \snort\bin: It is used to change the directory.
.\snort.exe -V -i 0: It starts snort as just a packet sniffer.

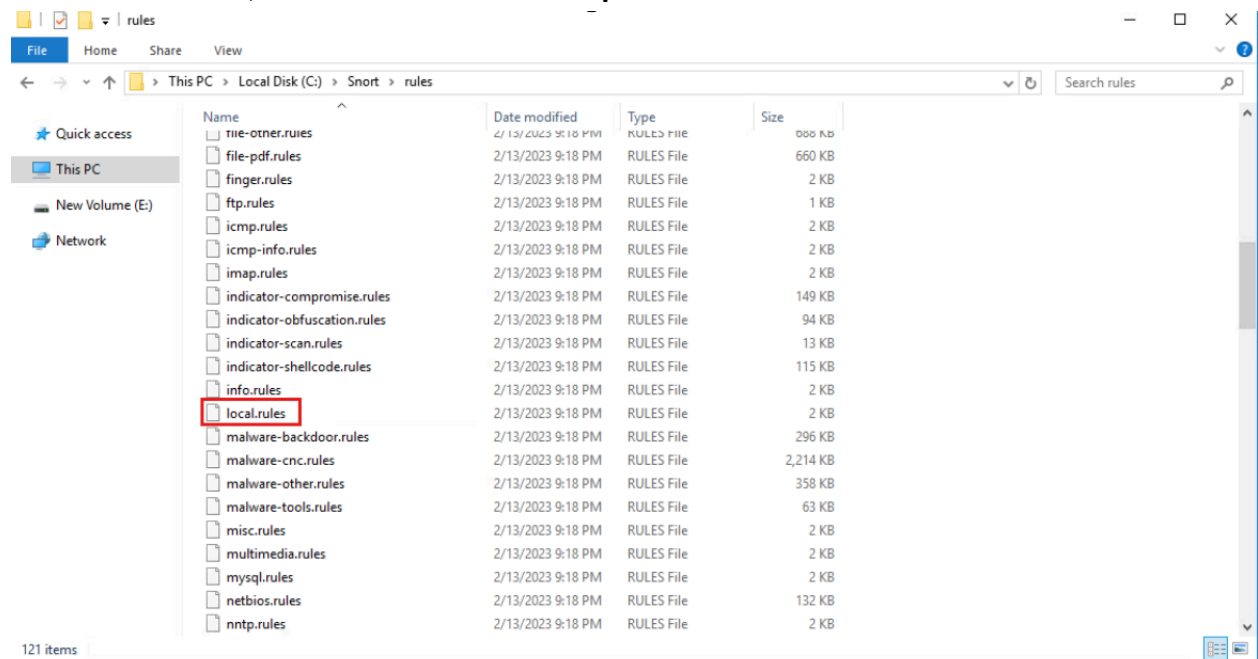
```
PS C:\Users\Administrator> cd \snort\bin
PS C:\snort\bin> .\snort.exe -V -i 0

-*)> Snort! <*-
o''~
...~
Version 2.9.11.1-WIN32 GRE (Build 268)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2017 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using PCRE version: 8.38 2015-11-23
Using ZLIB version: 1.2.3

PS C:\snort\bin>
```

9. From the taskbar, open **File Explorer**.

10. In the **File Explorer** window, navigate to **Local Disk (C:) > Snort > rules**, right-click **local.rules**, and select **Edit with Notepad++**.



```
snort.conf local.rules
1 # Copyright 2001-2018 Sourcefire, Inc. All Rights Reserved.
2 #
3 # This file contains (i) proprietary rules that were created, tested and certified by
4 # Sourcefire, Inc. (the "VRT Certified Rules") that are distributed under the VRT
5 # Certified Rules License Agreement (v 2.0), and (ii) rules that were created by
6 # Sourcefire and other third parties (the "GPL Rules") that are distributed under the
7 # GNU General Public License (GPL), v2.
8 #
9 # The VRT Certified Rules are owned by Sourcefire, Inc. The GPL Rules were created
10 # by Sourcefire and other third parties. The GPL Rules created by Sourcefire are
11 # owned by Sourcefire, Inc., and the GPL Rules not created by Sourcefire are owned by
12 # their respective creators. Please see http://www.snort.org/snort/snort-team/ for a
13 # list of third party owners and their respective copyrights.
14 #
15 # In order to determine what rules are VRT Certified Rules or GPL Rules, please refer
16 # to the VRT Certified Rules License Agreement (v2.0).
17 #
18 #-----
19 # LOCAL RULES
20 #-----
21
22
```

11. Add the following signatures rules at the end of the file **local.rules**:

```
alert icmp any any -> any any (msg: "Ping packets detected"; sid: 1000001; )
alert tcp any any -> any any (msg: "TCP packets detected"; sid: 1000002; )
alert udp any any -> any any (msg: "UDP packets detected"; sid: 1000003; )
```

alert: It allows snort to generate an alert when the set condition is met.

any: It allows snort to look at all sources.

->: It is the direction from source to destination

```
22 alert icmp any any -> any any (msg: "Ping packets detected"; sid: 1000001; )
23 alert tcp any any -> any any (msg: "TCP packets detected"; sid: 1000002; )
24 alert udp any any -> any any (msg: "UDP packets detected"; sid: 1000003; )
25
```

12. From the menu, navigate to **File > Save** to save all the changes.

13. From the taskbar, open **Windows PowerShell**.

14. In the **Administrator: Windows PowerShell** window, execute the following command:

```
.\snort.exe -i 0 -c C:\Snort\etc\snort.conf -T
```



```
Administrator: Windows PowerShell

2 byte states : 15
4 byte states : 0
Characters      : 205237
States         : 158187
Transitions    : 26158345
State Density   : 64.6%
Patterns       : 10247
Match States    : 10541
Memory (MB)    : 79.56
  Patterns     : 0.86
  Match Lists  : 1.43
DFA
  1 byte states : 1.45
  2 byte states : 75.57
  4 byte states : 0.00
-----
[ Number of patterns truncated to 20 bytes: 588 ]
pcap DAQ configured to passive.
The DAQ version does not support reload.
Acquiring network traffic from "0".

--- Initialization Complete ---

o'!~
...~
-#> Snort! <#-
Version 2.9.11.1-WIN32 GRE (Build 268)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2017 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using PCRE version: 8.38 2015-11-23
Using ZLIB version: 1.2.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.0 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>

Snort successfully validated the configuration!
Snort exiting
PS C:\snort\bin>
```

15. Close all windows.